DS Home Assignment:
Building a Problem Resolution System for HPC Logs

Objective

High performance computing (HPC) combines computing resources to achieve performance beyond a single computer, using either custom supercomputers or clusters of individual machines.

Your task is to build an intelligent system that not only analyzes HPC logs but creates an end-to-end workow for automated problem resolution. The goal is to reduce downtime by providing precise and actionable recommendations. The solution should be designed to handle evolving log formats and recommendations, making it "future-proof." While latency is not the primary concern, take into consideration that you want this to function in a real-world environment where cost and latency are important.

Problem Statement

- Analyze the provided HPC logs dataset to identify errors.
- Recommend specic actions to address these errors.
- Note that new logs and recommendations are introduced occasionally, so the system needs to be adaptable to new log templates, formats and recommended actions that are introduced over time.

Dataset

- Download HPC dataset
- The dataset contains structured HPC logs with elds such as timestamps, event IDs, and log messages.
- The logs may include dierent types of entries (e.g., normal operations, warnings, and errors). Your focus should be on identifying errors and providing actionable recommendations.

Core Requirements

Data Processing and Analysis

- Process structured HPC logs
- Describe how data ingestion will work (don't implement, just diagram)

Model integration and agentic workow

- Develop a pipeline for log analysis. Note that there may be multiple stages, from identifying the errors to providing recommendations, so be sure to use the right tool for the job.
- You should include an agent that can
    - Find errors and explain them
    - Query knowledge sources (for example - documentation, external APIs)
        - Suggest remediation steps, back them up with citations and reasons.

Note that you can register for access to LLMs at build.nvidia.com, but you can also run them locally or via another API service.

Submission

A link to a github repository including the les:

- `report`, A text/pdf/Markdown le including a detailed description of your thought process, your solution and suggested future improvements. ● `README.md`, a Markdown le including **exact** and precise instructions to build and execute your tool
- Any other source les needed to build

Please invite guy@mave.com to the repository.